

Reach the Target

Implicit Graphs, State-Space BFS · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★ Intermediate

Problem

An implicit graph: from an integer x you may move to $x+1$, $x-1$, or $x*2$, staying within $[0, \text{LIMIT}]$. Find the fewest moves to turn a start value into a target.

The states are never stored - their neighbours are generated by the rules - so BFS over this implicit graph yields the shortest number of moves.

What to Compute

- Steps: the minimum number of moves from S to T, or -1 if unreachable.
- Reachable: Yes if T can be reached, otherwise No.

Input / Output Format

Input: One line: LIMIT S T (search within $0..LIMIT$).

Output: Two lines: Steps and Reachable.

Examples

Example 1

Input: 10 2 9

Output:

Steps: 3
Reachable: Yes

Explanation: 2 doubles to 4, doubles to 8, then +1 reaches 9 - three moves, all within $[0,10]$.

Example 2

Input: 12 1 11

Output:

Steps: 5
Reachable: Yes

Explanation: A shortest route such as 1 to 2 to 3 to 6 to 12 to 11 (or an equally short one) takes five moves.

Constraints

- From x the moves are $x+1$, $x-1$ and $x*2$.
- Every intermediate value must stay within $[0, \text{LIMIT}]$.
- Each move costs one step; the start is step 0.

Sample Run

Sample Input

```
10 2 9
```

Sample Output

```
Steps: 3  
Reachable: Yes
```